Development of IGDI Assessments for 3-Year-Olds and Review of Literacy Interventions

Kayne Lussier, McNair Scholar
Dr. Scott McConnell and Kristin Schuster, IGDILab and CEHD

Introduction

Individual Growth and Development Indicators (IGDI) are brief and easy-to-use assessments to determine literacy and early language for 3 to 5-year-olds. The current research of IGDILab is to design specific assessments for 3-year-olds that focus on pre-literacy skills, such as alphabet knowledge and print knowledge. In utilizing these assessments, it is important to find methods to aid students in developing skills they may struggle with.

Purpose of Research: IGDILab has created assessments for 4-year-olds, but specific targeting for 3-year-olds will allow for clear developmental paths.

State of Research

Currently, the research is being coordinated with 7 sites in and around Minneapolis through the Urban and Outreach-Engagement Center (UROC). We are working with New Horizon Academies in Maple Grove, St. Paul, Prior Lake, Woodbury, Bloomington, and Chaska (Fig. 1). The current assessment formats are Picture Naming, Point to Picture (Fig. 2), Letter Naming, Letter Orientation, and Robot Blending. At this stage, the current goal is to determine the accuracy of the assessments with a large enough pool of participants (115).

After establishing the literacy proficiency of a student, the educators must be able to find evidence-based interventions. As APEL has yet to be designed for 3-year-olds, the literature reviews being conducted at IGDILab are integral at this stage.

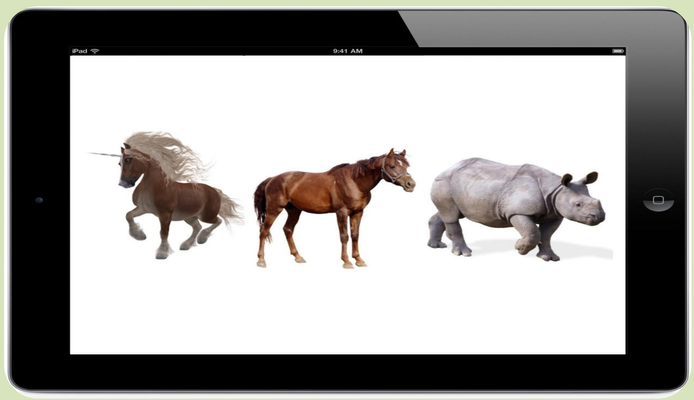


Figure 2. Example of Point to Picture, the prompt being "Point to the Rhino"

Dialogic Reading Interventions

Current research into interventions at this age is limited, but it has been growing in recent years.

Dialogic reading, a method that can either pair the student with another student or an educator, has long been utilized as a method of scaffolding for students, and a large amount of work has been centered around it.

Lonigan et al. (2013) conducted a study that looked at the skills of emergent literacy using dialogic reading interventions to create great growth. Howerver, combining the interventions in different skills saw diminished results.

As seen with research from Wasik et al. (2005) and Rahn et al. (2016), the intervention with the greatest traction is dialogic reading.

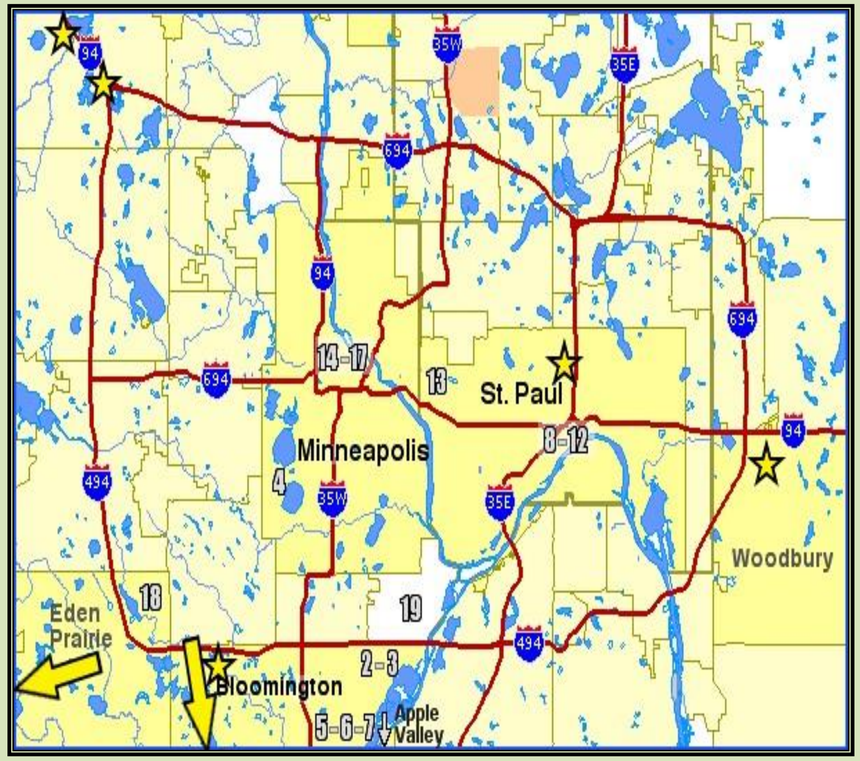


Figure 1. Map of the metro area, the stars and arrows representing our New Horizon sites.

Play Interventions

An important aspect of play interventions is the public perception surrounding the use of play in the education of young students. The literature is currently mixed in efficacy.

According to a study from Jung et al. (2016), even professionals in the field are unsure of play's ability to help children in their literacy development.

While Han et al. (2010) saw great gains in their study, Goodman and Dent (2017) saw no substantial differences between regular literacy interventions and the hybrid version with play.

Another important aspect is research into play assessments, which shows evidence of correlation between the quality of play in Head Start and later academic readiness once students reach elementary school (Bell et al., 2016).

Future Implications

Once the research and the literature reviews are completed for these particular IGDI assessments, we can begin work on early intervention. These assessments will be integrated with Automated Performance Evaluation of Early Literacy and Language (APEL), an application that can pair the literacy level of a student with reviewed interventions to set the child on track to grade level.

An issue with the current work is that the focus is only with New Horizon Academies, child care and early instruction that must be paid for. There are vouchers to aid lower-income families, but there is a skew in the children serviced, especially in the suburban neighborhoods these academies are located. Future work will be directed toward ensuring efficacy for all circumstances of living.

References

Bell, E. R., Greenfield, D. B., Bulotsky-Shearer, R. J., & Carter, T. M. (2016). Peer play as a context for identifying profiles of children and examining rates of growth in academic readiness for children enrolled in Head Start. *Journal of Educational Psychology*, *108*(5), 740–759. https://doi.org/10.1037/edu0000084

Han, M., Moore, N., Vukelich, C., & Buell, M. (2010). Does Play Make a Difference? How Play Intervention Affects the Vocabulary Learning of At-Risk Preschoolers. *American Journal of Play*, *3*(1), 82–105.

Jung, E., Zhang, Y., & Zhang, Y. (2017). Future professionals' perceptions of play and intended practices: the moderating role of efficacy beliefs. *Early Child Development and Care*, *187*(8), 1335–1348. https://doi.org/10.1080/03004430.2016.1169178

Lonigan, C. J., Purpura, D. J., Wilson, S. B., Walker, P. M., & Clancy-Menchetti, J. (2013). Evaluating the components of an emergent literacy intervention for preschool children at risk for reading difficulties. *Journal of Experimental Child Psychology*, *114*(1), 111–130. https://doi.org/10.1016/j.jecp.2012.08.010

Rahn, N. L., Coogle, C. G., & Storie, S. (2016). Preschool children's use of thematic vocabulary during dialogic reading and activity-based intervention. The Journal of Special Education, 50(2), 98-108.

Steen, R. L. (Ed.). (2017). *Emerging Research in Play Therapy, Child Counseling, and Consultation:* IGI Global. https://doi.org/10.4018/978-1-5225-2224-9

Wasik, B. A., Bond, M. A., & Hindman, A. (2006). The Effects of a Language and Literacy Intervention on Head Start Children and Teachers. Journal of Educational Psychology, 98(1), 63–74. https://doi.org/10.1037/0022-0663.98.1.63

TRIO RONALD E. MCNAIR POST-BACCALAUREATE ACHIEVEMENT PROGRAM

UNIVERSITY OF MINNESOTA

IGDILAB